$\mathscr{P}$

Problem Statement & Notation

Setting. A slider-crank chain OCP. The crank OC of radius r turns counter-clockwise at N rpm about the fixed centre O. The connecting rod CP has length $l = nr$, mass m_c, centre of mass G at l_G from the crank pin and radius of gyration k_G. The piston (bore D, reciprocating mass m_p) slides along the line of stroke. A rim-type flywheel of mean diameter D_f and a counterweight balancing the revolving mass plus a fraction c of the reciprocating mass are fitted.

Coordinates. θ is the crank angle from inner dead centre (TDC) and $\psi \in [0, 720^\circ)$ the four-stroke cycle angle (0 TDC suction, 180 BDC, 360 firing TDC, 540 BDC). Forces are positive toward the crank; torque T is positive in the direction of rotation.

Data (live).

$D = 86\,\text{mm}$	$r = 43\,\text{mm}$	$n = l/r = 3.40$	$l = 146.2\,\text{mm}$	$r_c = 10.0$	$m_p = 0.45\,\text{kg}$	$m_c = 0.55\,\text{kg}$	$l_G = 0.30\,l$
$k_G = 0.28\,l$	$N = 3{,}000\,\text{rpm}$ ($\omega = 314.2\,\text{rad/s}$)	axis: vertical	$p(\psi)$: Wiebe	load $= 80\,\%$	$p_1 = 1.00\,\text{bar}$	$\gamma = 1.300$	
$\theta_s = 20^\circ$	$\Delta\theta_b = 50^\circ$	$C_s = 2\,\%$	$D_f = 300\,\text{mm}$	$\rho = 7850\,\text{kg/m}^3$	$b/t = 2.0$	$c = 0.50$	

$r,\ l,\ n$	crank radius, rod length, obliquity ratio l/r
$\theta,\ \psi,\ \varphi$	crank angle, cycle angle, rod obliquity
$x,\ v,\ a$	piston displacement, velocity, acceleration
$\omega_c,\ \alpha_c$	angular velocity / acceleration of the rod
$F_g,\ F_I,\ F_P$	gas force, inertia force, piston effort
$F_Q,\ F_N$	thrust in rod, side thrust on cylinder wall
$F_T,\ F_R$	crank effort (tangential) and radial force
$T_c,\ T_w$	torque of the correction couple and of rod weight
$\Delta E,\ C_E,\ C_s$	max. fluctuation of energy; coefficients of fluctuation of energy and speed
$I,\ m_{\text{rim}},\ b,\ t$	flywheel inertia, rim mass, width, thickness
$m_R,\ m_{\text{rot}},\ c$	reciprocating and revolving masses, balance fraction

Assumptions.

- Rigid links, frictionless joints; crank balanced about O.
- Constant crank speed during the force analysis (the flywheel keeps the ripple within C_s).
- Gravity included; single-zone ideal gas with constant γ; instantaneous gas exchange at dead centres.
- Flywheel inertia from the rim alone (conservative).

Numerics. 1440 crank positions per cycle (0.5°), RK4 for the pressure equation, trapezoidal integrals.

References.

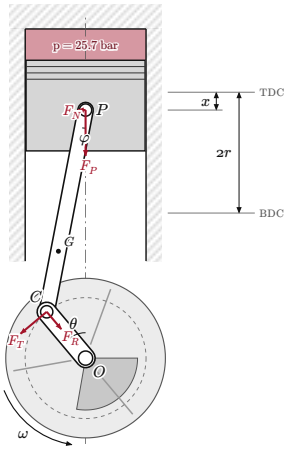
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Slider–Crank Mechanism of an IC Engine

Dynamic analysis of a single-cylinder four-stroke engine. The drawing is to scale and every number on this page is recalculated from first principles whenever a parameter changes.

Power stroke

$\psi = 400.0^\circ$, $\theta = 40.0^\circ$



O crank centre C crank pin P piston (gudgeon) pin G rod centre of mass θ crank angle, φ rod angle

red arrows: F_P piston effort, F_N side thrust, F_T crank effort, F_R radial force grey sector: counterweight

At this instant

Motion			Forces			Torque & speed		
crank angle	θ	40.0 °	cylinder pressure	p	25.68 bar	turning moment	T	407.6 N·m
piston displace…	x	12.70 mm	gas force	F_g	14.33 kN	mean (load) tor…	T_{mean}	39.21 N·m
piston velocity	v	10.68 m/s	piston inertia force	F_I	1.575 kN	energy above start	E	−164.7 J
piston acceleration	a	3,499 m/s²	piston effort	F_P	12.76 kN	speed with flyw…	N_{fw}	2990.7 rpm
rod angle	φ	10.90 °	force in rod	F_Q	12.93 kN			
rod angular velo…	ω_c	72.08 rad/s	side thrust	F_N	2.151 kN			
rod angular a…	α_c	−18,000 rad/s²	crank effort	F_T	9.479 kN			
			radial force	F_R	6.317 kN			

Results for the whole cycle

Power & torque	Flywheel	Pressure & piston motion	Forces
P_i 12.32 kW Indicated power $p_{mi} = 9.86$ bar	ΔE 517.5 J Max. fluctuation of energy $C_E = 1.05$	p_{max} 52.45 bar Peak cylinder pressure at $\psi = 373.5^\circ$	$F_{Q,max}$ 27.59 kN Max. force in connecting rod max. tension 2.47 kN
T_{mean} 39.21 N·m Mean turning moment $\eta_i = 48$ %	I 0.2621 kg·m² Flywheel moment of inertia for $C_s = 2$ %	v_{max} 14.08 m/s Peak piston speed mean 8.6 m/s	$\lvert F_N \rvert_{max}$ 2.576 kN Max. side thrust on wall crank pin 25.2 kN
T_{max} 491.6 N·m Peak turning moment at $\psi = 385.5^\circ$	m_{rim} 11.65 kg Flywheel rim mass $b \times t = 56.1 \times 28.1$ mm	a_{max} 5,492 m/s² Peak piston acceleration $= 560\ g$	$F_{s,max}$ 2.073 kN Max. shaking force on frame optimum $c = 0.60$

$\mathcal{K}$

Kinematics: Velocity & Acceleration

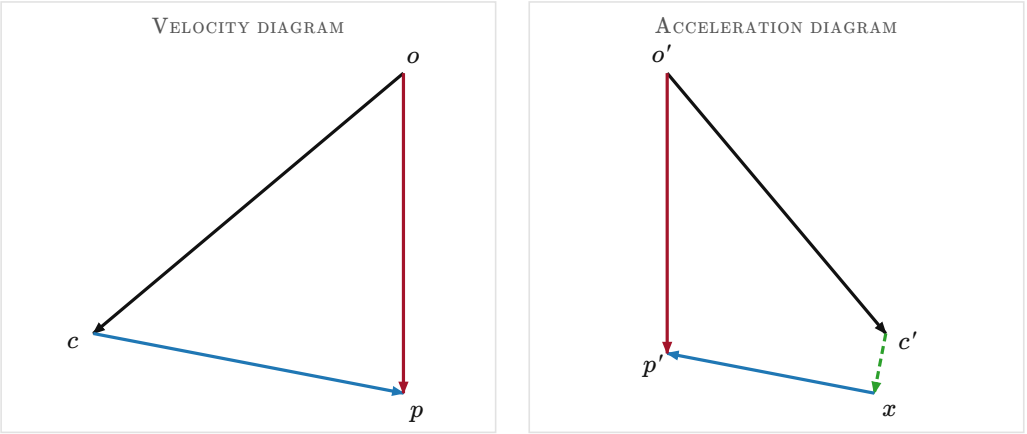
How fast the piston and rod move at every crank angle. These numbers drive all the inertia forces.

What the results say

- Peak piston speed $v_{\max} = \mathbf{14.1\ m/s}$ at $\theta = 75.0^\circ$, not at 90°: the inclined rod makes the piston move faster during the first half of the stroke.
- Acceleration at TDC ($\mathbf{5{,}492\ m/s^2}$, 560 g) is $(n{+}1)/(n{-}1) = \mathbf{1.83}$ times that at BDC, so the inertia force is largest at TDC.
- The simplified formula $a \approx r\omega^2(\cos\theta + \cos 2\theta/n)$ is off by at most **1.4 %** of $r\omega^2$ for $n = 3.40$; a longer rod makes it even closer.
- The rod swings $\pm 17.1^\circ$ about the line of stroke with peak angular acceleration $\mathbf{30{,}400\ rad/s^2}$.

Velocity and acceleration diagrams (relative velocity method)

Drawn to scale at the animation angle; they move with the mechanism. Vectors: oc crank-pin velocity $\perp OC$, cp velocity of P relative to C $\perp CP$, op piston velocity along the stroke. In the acceleration diagram, $c'x$ is the radial (centripetal) and xp' the tangential component of P relative to C.



Vector	Meaning	Magnitude
$v_C = r\omega$	crank-pin velocity (oc)	13.51 m/s
v_P	piston velocity (op)	10.68 m/s
$v_{PC} = \omega_c\, l$	P relative to C (cp)	10.54 m/s
$a_C = r\omega^2$	crank pin, centripetal (o′c′)	4,244 m/s²
$a^r_{PC} = \omega_c^2 l$	radial component (c′x)	759.6 m/s²
$a^t_{PC} = \alpha_c\, l$	tangential component (xp′)	2,632 m/s²
a_P	piston acceleration (o′p′)	3,499 m/s²

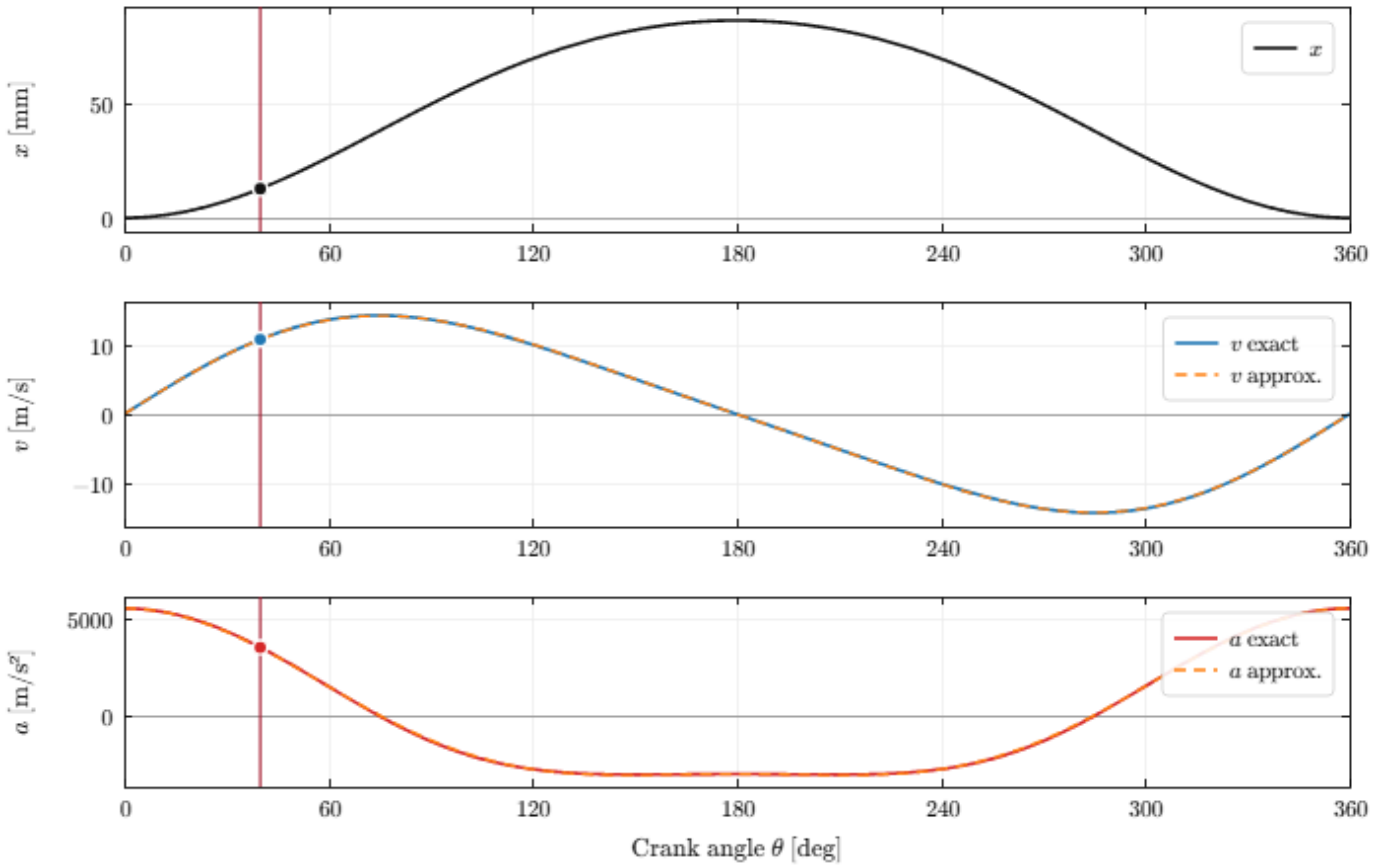


Fig. 2. Piston displacement, velocity and acceleration over one revolution (solid: exact, dashed: approximate formula).

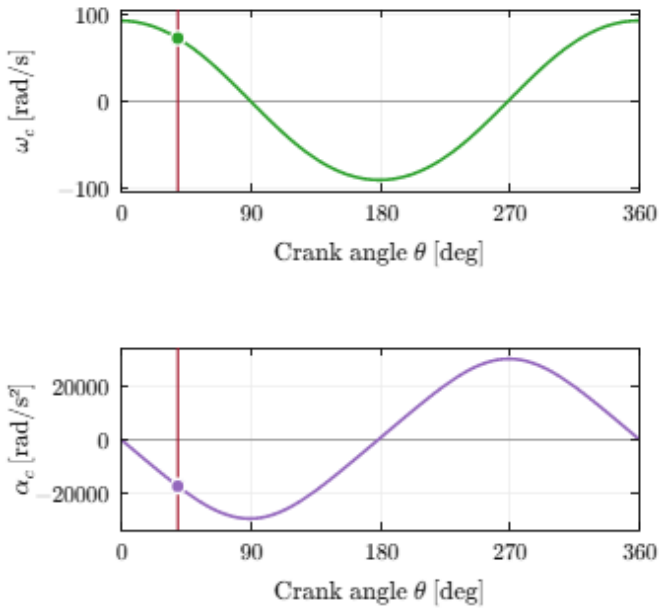


Fig. 3. Angular velocity and angular acceleration of the connecting rod.

How it is calculated

$$x = r\left[(1-\cos\theta) + n - \sqrt{n^2 - \sin^2\theta}\right], \qquad n = \frac{l}{r}, \qquad \omega = \frac{2\pi N}{60}$$

$$v = r\omega\left[\sin\theta + \frac{\sin 2\theta}{2\sqrt{n^2 - \sin^2\theta}}\right], \qquad a = r\omega^2\left[\cos\theta + \frac{n^2\cos 2\theta + \sin^4\theta}{\left(n^2 - \sin^2\theta\right)^{3/2}}\right]$$

$$\sin\varphi = \frac{\sin\theta}{n}, \qquad \omega_c = \frac{\omega\cos\theta}{\sqrt{n^2 - \sin^2\theta}}, \qquad \alpha_c = -\frac{\omega^2\sin\theta\,(n^2-1)}{\left(n^2 - \sin^2\theta\right)^{3/2}}$$

$$\text{approximate: } v \approx r\omega\left(\sin\theta + \tfrac{\sin 2\theta}{2n}\right), \qquad a \approx r\omega^2\left(\cos\theta + \tfrac{\cos 2\theta}{n}\right)$$

$$\omega = 314.2\ \text{rad/s} \qquad v_{\max} = 14.08\ \text{m/s} \qquad \bar{v}_p = \tfrac{4rN}{60} = 8.6\ \text{m/s} \qquad a_{\text{TDC}} = r\omega^2\left(1 + \tfrac{1}{n}\right) = 5{,}492\ \text{m/s}^2$$

$$a_{\text{BDC}} = -r\omega^2\left(1 - \tfrac{1}{n}\right) = -2{,}996\ \text{m/s}^2 \qquad \omega_{c,\max} = 92.4\ \text{rad/s} \qquad \alpha_{c,\max} = 30{,}370\ \text{rad/s}^2$$

$\mathcal{G}$

Cylinder Pressure & Indicator Diagram

The gas pushes the piston; this pressure is the input for the force analysis.

What the results say

- Peak pressure **52.5 bar** occurs 13.5° after firing TDC (for best efficiency it is usually 10–16° after TDC).
- Indicated work **492.7 J** per cycle gives $p_{\mathrm{mi}} = 9.86$ bar and $P_i =$ **12.3 kW** at 3,000 rpm (one power stroke every two revolutions).
- Indicated efficiency **48 %** against the ideal Otto value 49.9 %. The gap comes from the finite burn duration and the pumping loop.
- Pumping (gas-exchange) work: $(p_1 - p_{\mathrm{exh}})V_s = -3.19$ J per cycle.

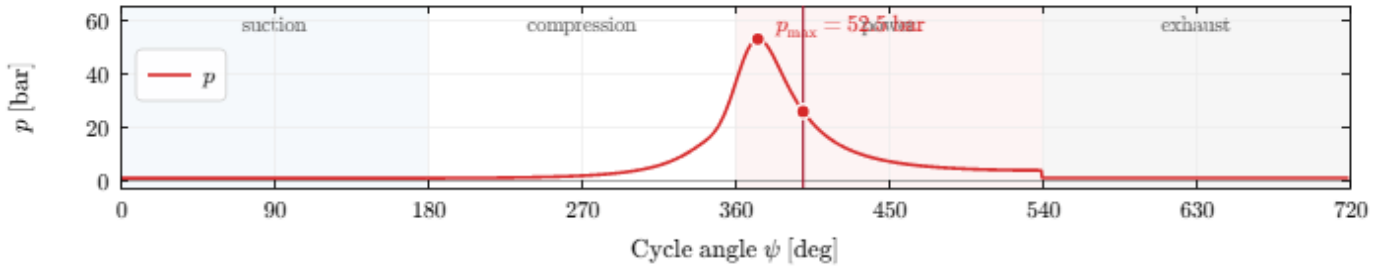


Fig. 4. Cylinder pressure over the four-stroke cycle.

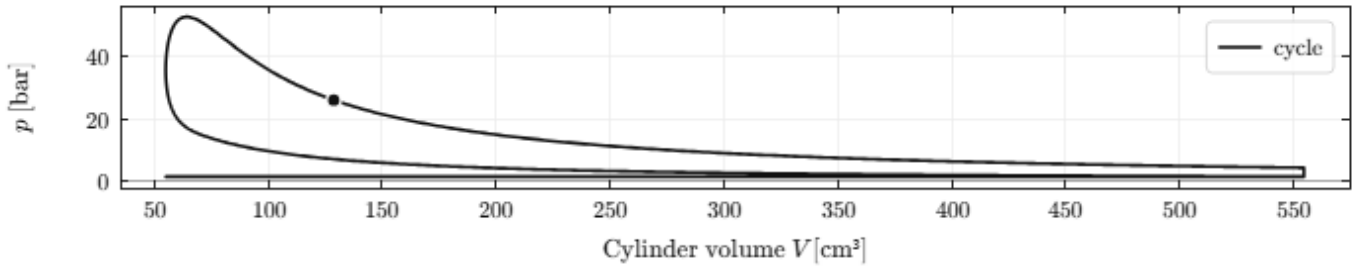


Fig. 5. Indicator (p–V) diagram. The enclosed area is the indicated work per cycle; the dot follows the animation.

How it is calculated

$$V(\theta) = V_c + \tfrac{\pi}{4}D^2\,x(\theta), \qquad V_s = \tfrac{\pi}{4}D^2\,(2r), \qquad V_c = \frac{V_s}{r_c - 1}$$

$$\frac{dp}{d\theta} = -\gamma\,\frac{p}{V}\,\frac{dV}{d\theta} + \frac{\gamma - 1}{V}\,\frac{dQ}{d\theta}, \qquad Q(\theta) = Q_{\mathrm{in}}\left\{1 - \exp\left[-5\left(\frac{\theta - \theta_0}{\Delta\theta_b}\right)^3\right]\right\}$$

$$W = \oint p\,dV, \qquad p_{\mathrm{mi}} = \frac{W}{V_s}, \qquad P_i = W\,\frac{N}{120}, \qquad \eta_i = \frac{W}{Q_{\mathrm{in}}}$$

First-law energy balance of the closed cylinder, integrated with RK4 (0.25° step). Combustion starts at $\theta_0 = 360^\circ - \theta_s$ and lasts $\Delta\theta_b$. Suction is at p_1, exhaust at $1.05\,p_{\mathrm{atm}}$, and the crank-case is at p_{atm}. Net heat released at full load is 2.6 kJ per litre of swept volume.

$V_s = 499.6\ \mathrm{cm}^3$ $V_c = 55.51\ \mathrm{cm}^3$ $Q_{\mathrm{in}} = 1{,}025\ \mathrm{J}$ $p_{\max} = 52.45\ \mathrm{bar}\ (\psi = 373.5^\circ)$ $W = \oint p\,dV = 492.7\ \mathrm{J}$

$p_{\mathrm{mi}} = 9.863\ \mathrm{bar}$ $P_i = 12.32\ \mathrm{kW}$ $\eta_i = 48\,\%\ \ (\eta_{\mathrm{Otto}} = 49.9\,\%)$

$\mathcal{F}$

Dynamic Force Analysis

Gas force minus the inertia of the moving parts gives the forces in every link and bearing.

What the results say

- The gas pushes with up to **29.9 kN**, but the rod carries only **27.6 kN**: near TDC the piston's own inertia absorbs part of the gas load.
- At $\psi = 0^\circ$ (suction) there is little gas pressure, so the inertia of the piston pulls on the rod: **tension 2.47 kN**. The rod and its bolts must be designed for this reversal.
- Largest side thrust **2.58 kN** at $\psi = 386^\circ$ (power). The cylinder wall wears most on this side.
- The crank-pin bearing load peaks at **25.2 kN**; the polar diagram shows in which direction, relative to the crank, the big-end bearing is loaded.

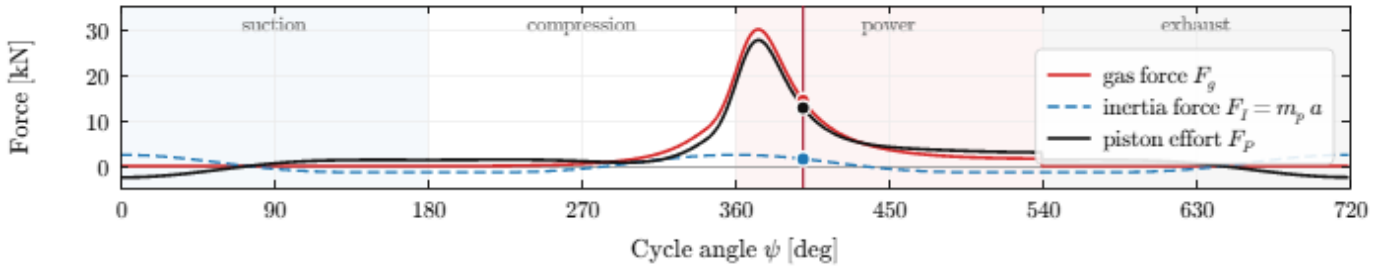


Fig. 6. Gas force, inertia force of the piston and net piston effort.

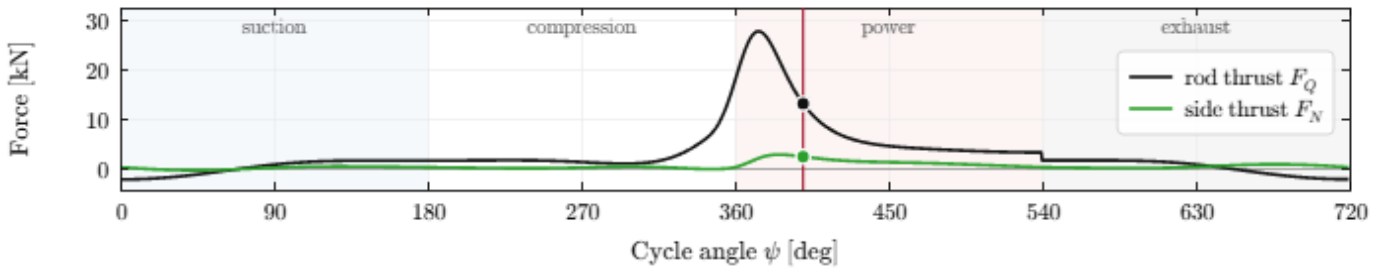
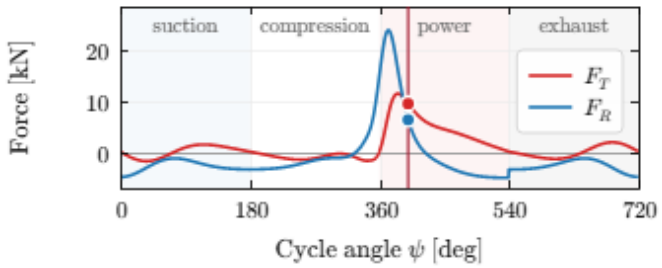


Fig. 7. Thrust in the connecting rod (+ compression, − tension) and side thrust on the cylinder wall.



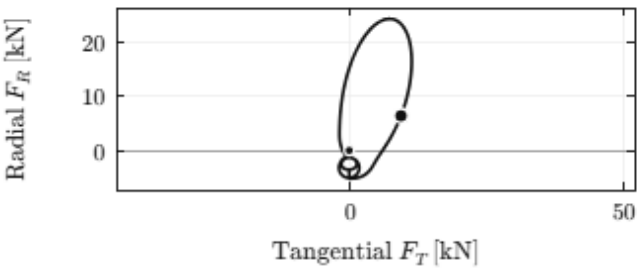


Fig. 8. Left: crank effort F_T and radial force F_R. Right: polar load diagram of the crank pin (in the crank frame), used to size the big-end bearing.

How it is calculated

$$F_g = (p - p_{\text{atm}})\,\tfrac{\pi}{4}D^2, \qquad F_P = F_g - m_R\,a\ (+\,m_R\,g \text{ if vertical}), \qquad m_R = m_p + \tfrac{l_G}{l}\,m_c$$

$$F_Q = \frac{F_P}{\cos\varphi}, \quad F_N = F_P\tan\varphi, \quad F_T = F_Q\sin(\theta+\varphi), \quad F_R = F_Q\cos(\theta+\varphi), \quad T_1 = F_T\,r$$

$$T = T_1 + T_c + T_w, \qquad T_c = \left(I_2 - I_G\right)\alpha_c\,\frac{\cos\theta}{\sqrt{n^2-\sin^2\theta}}, \qquad I_2 = m_c\,l_1(l-l_1)$$

The plots use the exact Newton–Euler solution of piston and rod (rod mass and moment of inertia included): $m_c\mathbf{a}_G = \mathbf{F}_C - \mathbf{F} + m_c\mathbf{g}$, $I_G\ddot{\beta} = (\mathbf{r}_C - \mathbf{r}_P)\times\mathbf{F} + (\mathbf{r}_C - \mathbf{r}_G)\times m_c(\mathbf{a}_G - \mathbf{g})$, $T = [\mathbf{r}_C \times (-\mathbf{F}_C)]_z$. The textbook two-mass method with correction couple T_c gives exactly the same torque; see the Verification tab.

$$F_{g,\max} = 29.88\ \text{kN} \qquad F_{Q,\max} = 27.59\ \text{kN (comp.)} \qquad F_{Q,\min} = -2.475\ \text{kN (tension)} \qquad |F_N|_{\max} = 2.576\ \text{kN}$$

$$F_{C,\max} = 25.18\ \text{kN (crank pin)} \qquad F_{G,\max} = 27.59\ \text{kN (gudgeon pin)}$$

$\mathcal{T}$

Turning-Moment Diagram & Flywheel

The engine torque swings above and below the steady load torque; the flywheel stores the surplus and gives it back.

What the results say

- The engine torque is negative for **50 %** of the cycle (suction, compression, exhaust); there the flywheel has to drive the piston.
- Speed is highest at $\psi = 522^\circ$ and lowest at $\psi = 362^\circ$. Between them the flywheel absorbs $\Delta E =$ **517.5 J**.
- $C_E = \Delta E / W =$ **1.05**, typical of a single-cylinder four-stroke engine; multi-cylinder engines are far smoother.
- Needed: $I =$ **0.262 kg·m²**, i.e. a rim of **11.7 kg** at 300 mm diameter. Because $I = \Delta E/(\omega^2 C_s)$, doubling the speed needs only about a quarter of the inertia.
- Rim speed 47.1 m/s is within the safe limit (hoop stress 17.4 MPa).

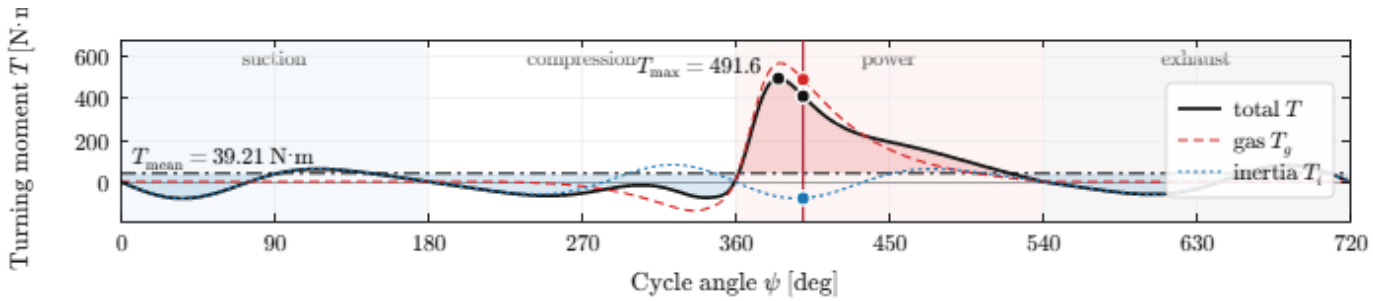


Fig. 9. Turning-moment diagram. Red areas: $T > T_{\text{mean}}$ (flywheel speeds up); blue: $T < T_{\text{mean}}$ (flywheel slows down).

Table of areas (textbook method)

Each loop is the area between T and T_{mean} between two crossing points. Taking the energy at the first crossing as E, the running total gives the energy at every crossing; the largest minus the smallest is ΔE.

Loop	from ψ	to ψ	Area (J)	Energy at end
start at point A		92.4°		E
A→B (above mean)	92.4°	146.4°	+12.69	$E + 12.69$
B→C (below mean)	146.4°	362.2°	−272.85	$E - 260.16$ **(min)**
C→D (above mean)	362.2°	521.3°	+517.47	$E + 257.31$ **(max)**
D→E (below mean)	521.3°	658.3°	−151.69	$E + 105.62$
E→F (above mean)	658.3°	708.1°	+22.67	$E + 128.29$
F→G (below mean)	708.1°	92.4°	−128.29	E (back to start)
Maximum fluctuation of energy $\Delta E = E_{\max} - E_{\min} =$ **517.47 J** (direct integration: 517.46 J)				

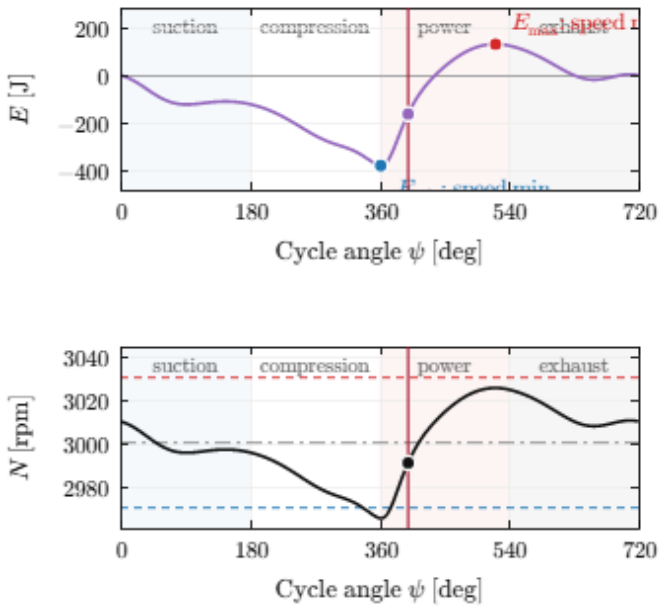


Fig. 10. Left: fluctuation of energy $E(\psi)$. Right: crankshaft speed with the designed flywheel, which stays inside the $\pm C_s/2$ band.

How it is calculated

$$T_{\text{mean}} = \frac{1}{4\pi}\int_0^{4\pi} T\,d\theta, \qquad E(\theta) = \int_0^{\theta} (T - T_{\text{mean}})\,d\theta, \qquad \Delta E = E_{\max} - E_{\min}$$

$$\Delta E = I\,\omega^2 C_s \;\Rightarrow\; I = \frac{\Delta E}{\omega^2 C_s}, \qquad C_E = \frac{\Delta E}{W}, \qquad m_{\text{rim}} = \frac{I}{R^2}, \qquad b\,t = \frac{m_{\text{rim}}}{2\pi R\,\rho}, \qquad \sigma_t = \rho\,v^2$$

$T_{\text{mean}} = 39.21\ \text{N m}$ $T_{\max} = 491.6\ \text{N m}\ (\psi = 385.5^\circ)$ $T_{\min} = -80.38\ \text{N m}$ $W = \int T\,d\theta = 492.7\ \text{J}$ $\Delta E = 517.5\ \text{J}$

$C_E = 1.05$ $I = 0.2621\ \text{kg m}^2$ $m_{\text{rim}} = 11.65\ \text{kg}$ $b \times t = 56.1 \times 28.1\ \text{mm}$ $v = \omega R = 47.1\ \text{m/s} \le 50\ \text{m/s (safe)}$

$\sigma_t = \rho v^2 = 17.4\ \text{MPa}$ $N_{\max}/N_{\min} = 3{,}025/2{,}965\ \text{rpm}$

$\mathcal{B}$

Balancing of Reciprocating Masses

The moving parts shake the engine frame; a counterweight on the crank can only partly cancel this.

What the results say

- Without balancing the reciprocating mass ($c = 0$), the frame is shaken along the stroke by up to **3.38 kN**.
- With $c = 0.50$ the peak becomes **2.07 kN** ($-39\ \%$), but a new force $c\, m_R\, \omega^2 r = 1.31$ kN appears across the stroke.
- Best compromise: $c^* = \mathbf{0.60}$, giving 1.81 kN. The secondary force (0.768 kN, at twice crank speed) cannot be balanced by a crank counterweight.

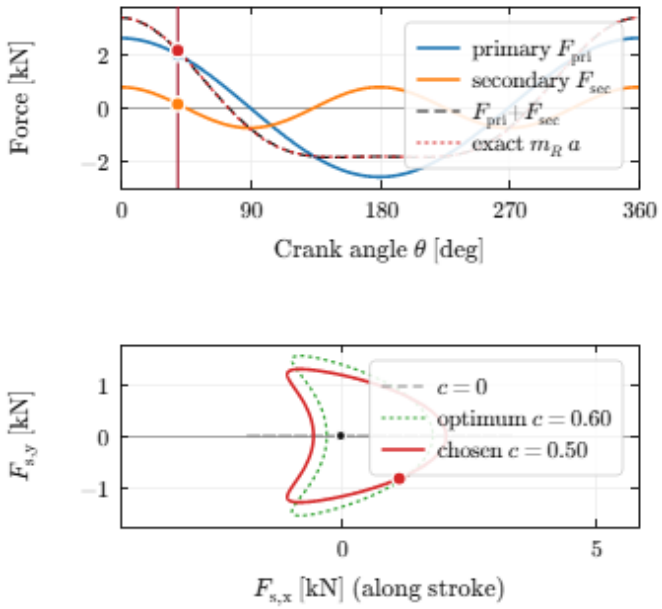


Fig. 11. Left: primary and secondary unbalanced forces over one revolution. Right: shaking force on the frame for $c = 0$, the chosen c, and the optimum c.

How it is calculated

$$m_R = m_p + \frac{l_G}{l}\, m_c, \qquad m_{\text{rot}} = \left(1 - \frac{l_G}{l}\right) m_c, \qquad m_B = m_{\text{rot}} + c\, m_R \ \text{(counterweight at radius } r\text{)}$$

$$F_{\text{pri}} = m_R\, \omega^2 r \cos\theta, \qquad F_{\text{sec}} = m_R\, \omega^2 r\, \frac{\cos 2\theta}{n}, \qquad \mathbf{F}_s = -\left(m_p\, \mathbf{a}_P + m_c\, \mathbf{a}_G + m_B\, \mathbf{a}_B\right)$$

$$\text{residual primary: } (1 - c)\, m_R \omega^2 r \cos\theta \text{ along the stroke}, \qquad c\, m_R \omega^2 r \sin\theta \text{ across it}$$

$$m_R = 0.615 \text{ kg} \qquad m_{\text{rot}} = 0.385 \text{ kg} \qquad m_B = 0.6925 \text{ kg at } r \qquad F_{\text{pri,max}} = 2.61 \text{ kN} \qquad F_{\text{sec,max}} = 0.7677 \text{ kN}$$

$$F_{s,\max}(c{=}0) = 3.378 \text{ kN} \qquad F_{s,\max}(c{=}0.50) = 2.073 \text{ kN} \qquad c^* = 0.60 \ \Rightarrow \ F_{s,\max} = 1.812 \text{ kN}$$

$\mathcal{W}$

Step-by-Step Solution at One Crank Angle

The standard textbook procedure (two-mass equivalent rod with correction couple), with your numbers substituted at every step.

1 Angular speed of the crank

$$\omega = \frac{2\pi N}{60} = \frac{2\pi \times 3{,}000}{60} = 314.2 \text{ rad/s}$$

2 Crank position and gas pressure (power stroke)

$\psi = 420.0^\circ \Rightarrow \theta = 60.0^\circ$ from TDC, $p = 13.8$ bar (abs.), from the pressure model

3 Obliquity of the connecting rod

$$n = \frac{l}{r} = \frac{146.2}{43} = 3.400, \qquad \sin\varphi = \frac{\sin\theta}{n} = \frac{0.866}{3.400} = 0.2547 \Rightarrow \varphi = 14.76^\circ$$

4 Piston displacement from TDC

$$x = r\left[(1-\cos\theta) + n - \sqrt{n^2 - \sin^2\theta}\right] = 43\left[(1-0.5) + 3.400 - 3.288\right] = 26.32 \text{ mm}$$

5 Piston velocity

$$v = r\omega\left[\sin\theta + \frac{\sin 2\theta}{2\sqrt{n^2 - \sin^2\theta}}\right] = 0.043 \times 314.2\left[0.866 + \frac{0.866}{2 \times 3.288}\right] = 13.48 \text{ m/s}$$

6 Piston acceleration (exact)

$$a = r\omega^2\left[\cos\theta + \frac{n^2\cos 2\theta + \sin^4\theta}{(n^2 - \sin^2\theta)^{3/2}}\right] = 0.043 \times 314.2^2\left[0.5 + (-0.1468)\right] = 1{,}499 \text{ m/s}^2$$

The usual approximate formula gives almost the same value:

$$a \approx r\omega^2\left(\cos\theta + \frac{\cos 2\theta}{n}\right) = 1{,}498 \text{ m/s}^2 \quad (\text{difference } 0.026\,\% \text{ of } r\omega^2)$$

7 Equivalent reciprocating mass (rod split into two point masses with the same CG)

$$m_R = m_p + \frac{l_G}{l} m_c = 0.45 + 0.30 \times 0.55 = 0.615 \text{ kg}, \qquad m_2 = m_c - \frac{l_G}{l} m_c = 0.385 \text{ kg at the crank pin}$$

8 Force due to gas pressure

$$F_g = (p - p_{\text{atm}})\frac{\pi D^2}{4} = (13.8 - 1.013) \times 10^5 \times \frac{\pi \times 0.086^2}{4} = 7{,}428 \text{ N}$$

9 Inertia force of the reciprocating parts

$$F_I = m_R\, a = 0.615 \times 1{,}499 = 921.9 \text{ N}$$

10 Net force on the piston (piston effort) (vertical engine: weight helps on the down-stroke)

$$F_P = F_g - F_I + m_R g = 7{,}428 - 921.9 + 6.031 = 6{,}512 \text{ N}$$

11 **Thrust in the connecting rod**

$$F_Q = \frac{F_P}{\cos\varphi} = \frac{6{,}512}{\cos 14.76^\circ} = 6{,}735\ \text{N}$$

12 **Side thrust on the cylinder wall**

$$F_N = F_P \tan\varphi = 6{,}512 \times \tan 14.76^\circ = 1{,}715\ \text{N}$$

13 **Crank effort and radial force at the crank pin**

$$F_T = F_Q \sin(\theta+\varphi) = 6{,}735 \sin 74.76^\circ = 6{,}498\ \text{N}, \qquad F_R = F_Q \cos(\theta+\varphi) = 1{,}771\ \text{N}$$

14 **Turning moment of the two-mass system**

$$T_1 = F_T \times r = 6{,}498 \times 0.043 = 279.4\ \text{N m}$$

15 **Correction couple for the rod's moment of inertia** (the two point masses do not have the real rod's inertia)

$$l_1 = l - l_G = 102.3\ \text{mm}, \quad L = l_1 + \frac{k_G^2}{l_1} = 118.7\ \text{mm}, \quad \alpha_c = -25{,}400\ \text{rad/s}^2, \quad T' = m_c\, l_1\, (l - L)\, \alpha_c = -39.29\ \text{N m}$$

Torque this couple puts on the crankshaft:

$$T_c = T' \frac{\cos\theta}{\sqrt{n^2 - \sin^2\theta}} = (-39.29) \times \frac{0.5}{3.288} = -5.975\ \text{N m}$$

16 **Torque due to the weight of the rod** (crank-pin share; vertical engine)

$$T_w = m_2\, g\, r \sin\theta = 0.385 \times 9.81 \times 0.043 \times 0.866 = 0.1406\ \text{N m}$$

17 **Net turning moment on the crankshaft**

$$T = T_1 + T_c + T_w = 279.4 + (-5.975) + 0.1406 = \boxed{273.6\ \text{N m}}$$

exact Newton–Euler solver: $T = 273.6$ N m difference $= 0\,\%$ $T - T_{\text{mean}} = 234.4$ N m (flywheel speeding up)

S

What Changes When a Parameter Changes?

A bar of +0.8 means a 1 % increase in that parameter raises the chosen result by about 0.8 %; negative bars mean it falls.

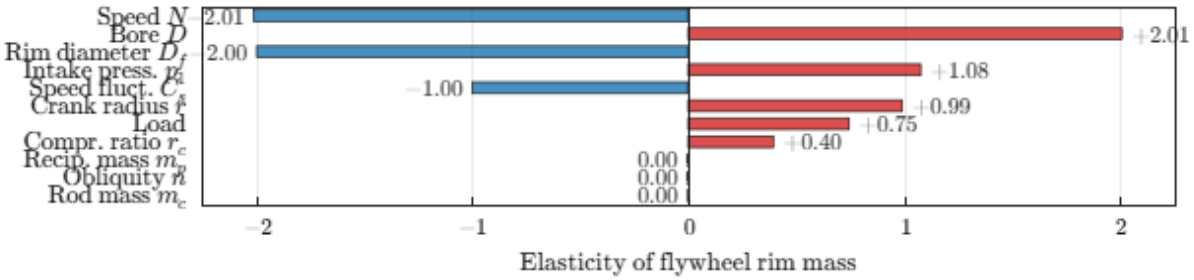


Fig. 12. Elasticity $\partial \ln y / \partial \ln x$ of the selected result to each input at the current design (central differences, ±1 %).

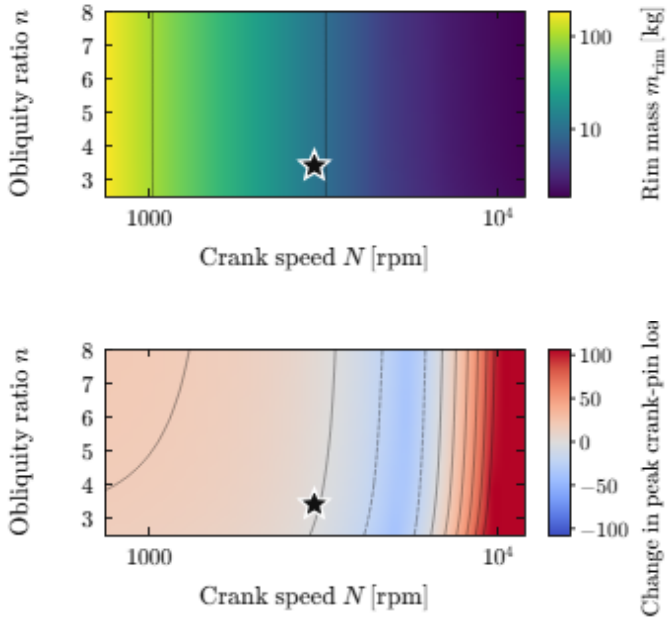


Fig. 13. Design space over crank speed N and obliquity ratio n. Left: flywheel rim mass needed for the chosen C_s. Right: change of peak crank-pin load relative to the current design (★). Click a map to move the design there.

$\mathcal{V}$

Verification of the Calculations

Every result is cross-checked against an independent method.

Benchmark A: textbook problem. A horizontal engine has $r = 300$ mm, $l = 1.2$ m, $D = 0.5$ m, $m_R = 250$ kg and $N = 250$ rpm, with a net pressure of 0.35 N/mm² at $\theta = 60°$ from IDC. It is solved by the hand-calculation chain (approximate acceleration) and by the exact solver.

Quantity ($\theta = 60°$)	Hand calculation	Exact solver	Difference
Inertia force F_I	19.277 kN	19.282 kN	0.030 %
Piston effort F_P	49.446 kN	49.440 kN	0.012 %
Thrust on side of cylinder F_N	10.965 kN	10.964 kN	0.012 %
Thrust in connecting rod F_Q	50.647 kN	50.641 kN	0.012 %
Crank effort F_T	48.304 kN	48.298 kN	0.012 %
Turning moment T	14.491 kN·m	14.490 kN·m	0.012 %

Benchmark B: air-standard Otto cycle. For $r_c = 8,\ \gamma = 1.4$ the integrated cycle must give $\eta = 1 - r_c^{1-\gamma}$.

Quantity	Numerical cycle	Closed form	Rel. error	
Thermal efficiency η	0.564722	0.564725	5.2e-6	✓ pass

Live checks on the current design (recomputed on every change):

Check	Compared	Rel. difference	
Crank torque: Newton–Euler vs energy (virtual-power) method	max $\lvert\Delta T\rvert$ / max $\lvert T\rvert$	3.5e-16	✓ pass
Crank torque: textbook two-mass rod + correction couple vs Newton–Euler	max $\lvert\Delta T\rvert$ / max $\lvert T\rvert$	9.2e-16	✓ pass
Cycle work: $\int T\,\mathrm{d}\theta$ (mechanism) vs $\oint p\,\mathrm{d}V$ (thermodynamics)	492.698 J vs 492.690 J	1.7e-5	✓ pass
Net work of all inertia forces over one cycle	$\oint T_i\,\mathrm{d}\theta$ / $\oint \lvert T_i\rvert\,\mathrm{d}\theta$	6.9e-18	✓ pass
Piston velocity: closed form vs finite difference of $x(\theta)$	max error / $r\omega$	2.4e-9	✓ pass
Piston acceleration: closed form vs finite difference	max error / $r\omega^2$	1.7e-7	✓ pass
Speed fluctuation with designed flywheel = requested C_s	2.0000 % vs 2.0000 %	0.0e+0	✓ pass

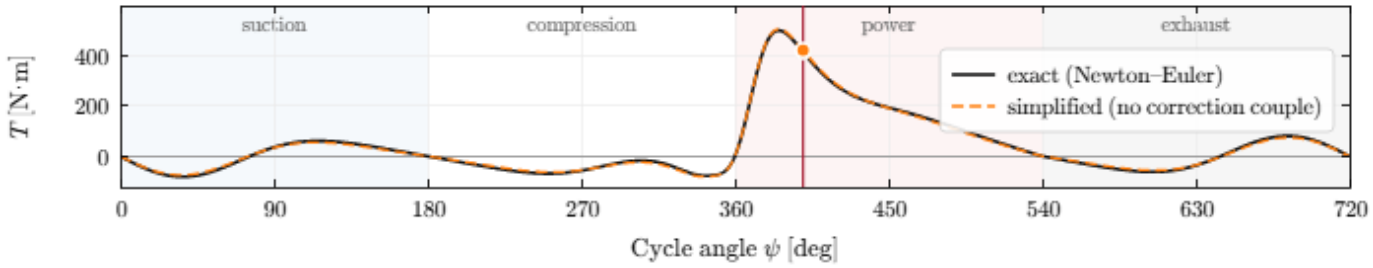


Fig. 14. Exact turning moment against the common simplified method (two-mass rod, approximate acceleration, no correction couple). The gap is the error of that simplification for this design.